



Connecting Trellit Voice Agents via Twilio to 3CX for Batch Call Capabilities

Main Takeaway

It is technically feasible to connect Trellit voice agents to a client's 3CX PBX system for outbound batch call capabilities using Twilio SIP trunking. Twilio functions as the bridge, allowing programmable voice agents (such as those built with Trellit) to initiate calls via its Programmable Voice or SIP trunk service, which can then route batches of calls to the client's 3CX PBX environment for further handling or distribution.

Technical Workflow

1. Voice Agent Call Origination (Trellit)

- Trellit acts as the programmable voice agent platform, handling API-driven call logic and contact batching.
- Batch outbound calls are triggered via Trellit's workflow automation.

2. Twilio SIP Trunk as Call Relay

- Trellit uses Twilio's Programmable Voice API or direct SIP trunk registration to initiate outbound calls.
- Calls are routed from Trellit to the Twilio SIP trunk endpoint corresponding to the client's dedicated number.
- Twilio acts as the central call switch, handling outbound call signaling.

3. 3CX PBX Integration

- The 3CX PBX is set up to recognize Twilio as a SIP trunk provider.
- SIP trunk configuration is completed in 3CX with registrar/server details provided by Twilio (typically in the format `xxxx.pstn.twilio.com` and port 5060).

3CX Admin Console

Dashboard Users Voice & Chat Phones Outbound Rules Departments Office Hours Call Handling Reports Event Log Recordings Backup Integrations Advanced System

Admin Apps

Add Trunk Save

General DID Numbers SMS Options

Name* Mobile Trunk

Default route User 100 admin

☐ Create an outbound rule for this SIP Trunk *Untick option*

Voip Provider

Country Worldwide Provider Twilio

Follow the Twilio [Configuration Guide](#). Use this IP for Whitelisting 54.206.249.67

Account Details

Main Trunk Number* +614839679XX

⚠ DID numbers must be in e164 format: If your number is (813) 591-0130 enter it as +18135910130.

Authentication ID (SIP User ID)* Authentication password*

Server Details

Registrar/Server* test.pstn.twilio.com Port* 5060 ☐ Auto discovery

SIP trunk configuration for Twilio in 3CX Admin Console showing key settings like trunk name, provider, and server details.

- Outbound rules can be tailored to control batch call routing and processing within the 3CX dashboard.

General

Rule Name

E.164

Apply this rule to these calls

Calls to numbers starting with prefix

+

Calls from extension(s)

Calls from extension(s)

Calls to Numbers with a length of

Calls to Numbers with a length of

Calls from extension group(s)

+ Add

Make outbound calls on

Configure up to 5 backup routes for outgoing calls. Each route can be configured differently

			Strip Digits	Prepend
Route	1	Twilio	0	
Route	2	BLOCK CALLS	0	
Route	3	BLOCK CALLS	0	
Route	4	BLOCK CALLS	0	
Route	5	BLOCK CALLS	0	

3CX outbound call rule configuration showing Twilio as the primary route for calls with E.164 dialing format.

- 3CX then manages the receipt and distribution of calls, including call flow logic and endpoint connection.

Configuration Steps

Twilio SIP Trunk Setup in 3CX:

- Add SIP trunk in 3CX admin console.
- Provider: Twilio
- Country: Worldwide
- Main Trunk Number: Client's Twilio-provisioned number
- Registrar/Server: Provided by Twilio (e.g., test.pstn.twilio.com)
- Port: 5060
- Optionally, adjust outbound rules for batch calling and number formatting using E.164 format.

Batch Call Workflow:

- Trellit initiates multiple outbound calls using Twilio's API or SIP endpoint, associating calls with client number.

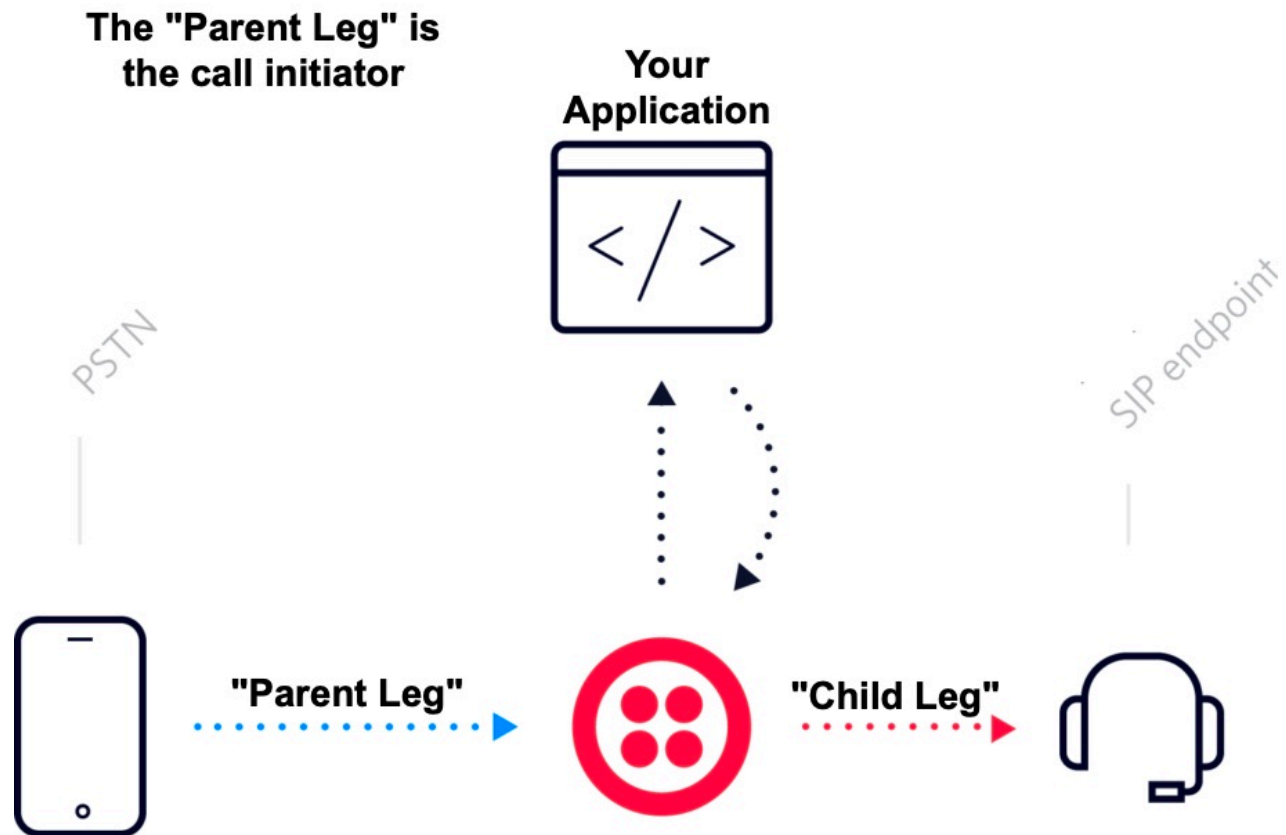


Diagram illustrating call legs and application interaction in a Twilio-based voice call setup between PSTN and SIP endpoints.

- Twilio relays all batch calls to assigned 3CX SIP trunk, treating each as a new call leg (originator → Twilio → 3CX PBX).
- 3CX further manages call flow, logging, and advanced routing per client's needs.

Reference Illustrations:

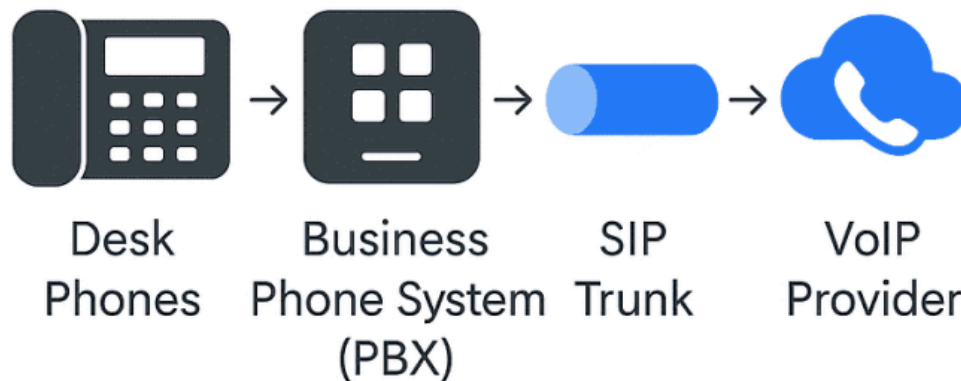
- The network diagrams below illustrate typical SIP trunk call flows and 3CX trunk setup. These serve as technical guides:
 - Twilio SIP trunk configuration within 3CX PBX
 - SIP call leg management from an API-programmable agent to Twilio SIP endpoint, then onward to PBX
 - Conceptual overview of SIP trunking integration for batch calling

SIP Trunking

DEFINITION

SIP trunking is a technology that allows businesses to make and receive phone calls over the internet using SIP (Session Initiation Protocol) instead of traditional telephone lines.

FUNCTIONALITY



SIP trunking workflow from desk phones through business phone system and SIP trunk to VoIP provider

Known Limitations and Considerations

- **Call Volume/Bursting:** Twilio SIP trunk and 3CX licensing or system limits may restrict max simultaneous calls; batch size may need throttling depending on licenses and hardware.
- **Number Masking:** Calls will show as originating from the Twilio-provisioned number unless specific number masking is enabled or permitted by 3CX and Twilio settings.
- **Quality/Reliability:** Both Twilio and 3CX must be correctly configured for NAT traversal, network whitelisting, and codecs to avoid call quality issues.

- **Advanced Workflow Logic:** Any complex pre-call logic, such as dynamic prompts or on-the-fly transfers, should be programmed into the Trellit agent environment before call hand-off.
- **Compliance:** Ensure compliance with local telephony and spam regulations for batch outbound calling.

Summary

Connecting Trellit voice agent workflows to a client's 3CX PBX for batch outbound calling via Twilio SIP trunk is achievable with proper configuration and API orchestration. Twilio mediates programmable call initiation and acts as the SIP trunk provider; 3CX configures Twilio as an inbound trunk, takes the calls, and manages further workflow. All steps can be customized for batch call automation, routing, and endpoint logic based on client requirements. Attention should be paid to call limits, number presentation, and regulatory considerations for batch dialing.

Reference Diagrams

Twilio SIP Trunk Setup in 3CX:

Twilio API Call Leg Model (outbound to 3CX via SIP):

SIP Trunking Explained (Batch call concept):

Outbound Call Rule Example (3CX):